The molecular weights of three globular polypeptides are shown in the following table.

Identity	Molecular weight (kDa)
Polypeptide **X**	30
Polypeptide **Y**	50
Polypeptide **Z**	70

Polypeptide **X** is connected to polypeptide **Y** through a disulfide bridge; otherwise, none of the polypeptides display any quaternary interactions.

When analyzed biochemically, polypeptide **X** will be detected in the fastest-migrating protein when performing:

I. size-exclusion chromatography with a reducing agent.

II. size-exclusion chromatography without any reducing agents.

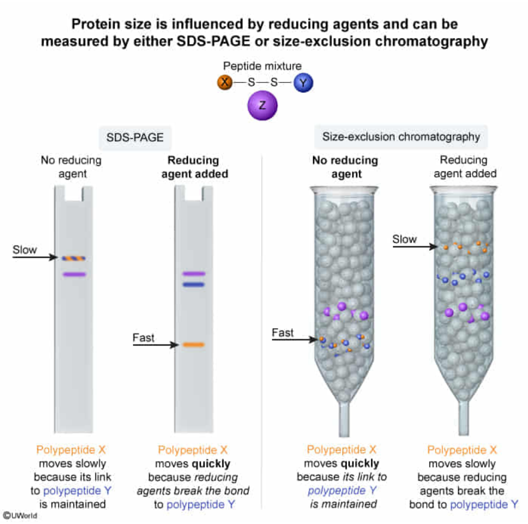
III. SDS-PAGE with a reducing agent.

IV. SDS-PAGE without any reducing agents.

- A. I only [10%]
- B. III only [37%]
- C. I and IV only [14%]
- ✔ D. II and III only [37%]

Correct
37% answered correctly

Explanation:



Two common techniques to analyze proteins are sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS-PAGE) and size-exclusion chromatography (SEC). Both techniques separate proteins by size, but the relationship between size and speed of migration differs between the two.

In SDS-PAGE, SDS coats the proteins in negative charges and then a strong electromotive force pulls the proteins through the gel matrix. Smaller proteins navigate the matrix more easily whereas larger proteins are hindered by the gel; thus, **smaller proteins migrate fastest in SDS-PAGE**. In contrast, SEC relies on gentler pushing forces like gravity flow, which allows molecules to diffuse into and out of the porous chromatography resin. Smaller proteins diffuse into the pores more readily, slowing them down, whereas larger proteins are excluded from the pores and move continually downward. Thus, **larger proteins migrate fastest in SEC**.

Both SDS-PAGE and SEC can be performed with or without reducing agents. In protein biochemistry, reducing agents are used to reduce disulfide bonds. With reducing agents, proteins run at their monomeric sizes; without reducing agents, disulfide bonds may cause dimerization of proteins, resulting in larger apparent sizes.

Polypeptide **X** has the smallest monomeric molecular weight of 30 kDa. Therefore, polypeptide **X** would be expected to migrate the fastest in SDS-PAGE, but only if a reducing is added to ensure it runs as a monomer **(Number III)**. Without a reducing agent, polypeptide **X** dimerizes with polypeptide **Y** to form an 80 kDa dimer. This is larger than any other protein in the mixture, and thus it migrates fastest in SEC **(Number II)**.

(Number I) With a reducing agent, dimers fall apart and polypeptide **X** becomes the smallest peptide in the mixture. Small peptides migrate *slowly* in SEC.

(Number IV) Without a reducing agent, polypeptide **X** dimerizes with polypeptide **Y** through a disulfide bond, forming the largest protein in the mixture. Large proteins migrate *slowly* in SDS-PAGE.

Educational objective:
Small proteins migrate quickly in SDS-PAGE but slowly in SEC whereas large proteins do the opposite. Disulfide bonds may cause some proteins to run as dimers, but reducing agents allow them to run at their monomeric sizes.

Time spent:0 sec
Subject:
Biochemistry

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System:
Amino Acids and Proteins